Parsing Command Lines

pythonProject_test Version control commandline_example

Project

pythonProject_test C:\Users\stomlinson\PycharmProjects\pytho
- .venv library root
  - Lib
  - Scripts
  - .gitignore
  - pyvenv.cfg
- commandline_example.py
- External Libraries
- Scratches and Consoles

commandline_example.py

```python
#Called with parameters "AAAAA" "TTTTT"  46


#Build a modified SmithWaterman.py Python programme
import argparse

parser = argparse.ArgumentParser(description='Aligning sequences...')
parser.add_argument( *name_or_flags: 'seq1',action="store",help="First sequence")
parser.add_argument( *name_or_flags: 'seq2',action="store",help="Second sequence")
parser.add_argument( *name_or_flags: 'anum',action="store",help="A number",type =int)

args = parser.parse_args()
print(args)
print("This is the first parameter", args.seq1)
print("This is the second parameter", args.seq2)
print("This is the third parameter", args.anum)
```

Run commandline_example

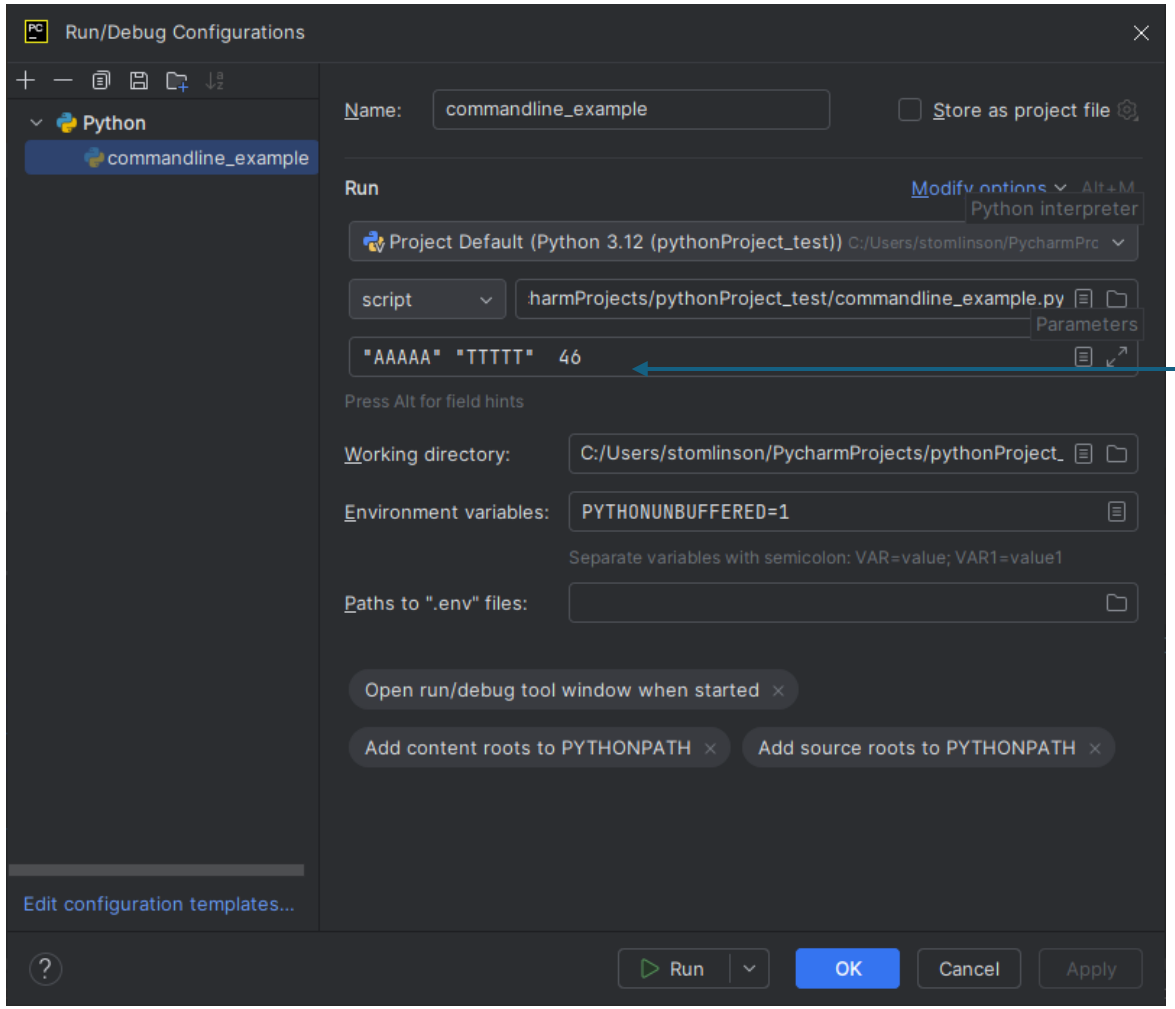
```
C:\Users\stomlinson\PycharmProjects\pythonProject_test\.venv\Scripts\python.exe C:\Users\stomlinson\PycharmProjects\pythonProject_test\commandline_example.py
Namespace(seq1='AAAAA', seq2='TTTTT', anum=46)
This is the first parameter AAAAA
This is the second parameter TTTTT
This is the third parameter 46

Process finished with exit code 0
```

pythonProject_test > commandline_example.py 17:7 CRLF UTF-8 4 spaces Python 3.12 (pythonProject_test)

Set args from here

- Note how the arguments fed to the programme are picked up when it runs

Command Line Arguments



Here

Why bother?

- The programme might be run on the command line (outside the IDE as)

```
python commandline_example.py "AAAAA" "TTTTT"  46
```

- So we can write a programme and then pass parameters to the programme from the command line
- We can even have the programme run as a standalone executable and then it behaved like a small programme performing one step in a longer pipeline
- Command lines are important for managing IO and parameters when running programmes standalone
- We could even write a web interface and this itself would then set the parameters and call the python script- results would be returned to the user.